MANAV ACHARYA

FULL STACK ENGINEER | VOICE AI & AGENT SYSTEMS

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Full Stack Engineer specializing in production voice AI and agent systems. Built low-latency, tool-calling workflows spanning telephony, RAG, CRM automation, and LLM-powered sales intelligence in TypeScript/Node.js.

EXPERIENCE

Alebex | *Full Stack Engineer* — Vancouver, BC | **Jun 2026 - Present**

- Built a **TypeScript/JavaScript CRM and sales-engagement layer** for an agentic voice platform, tracking lead lifecycle, dispositions, activity, and conversion outcomes.
- Integrated **Nylas email/calendar APIs with OAuth 2.0** for scheduling, availability flows, and CRM activity synchronization.
- Implemented **Twilio Voice/SMS browser telephony** with **WebRTC/PSTN bridging**, call routing, and automated interaction logging.
- Built an **OpenAI-powered RAG pipeline** for persistent lead memory, interaction summarization, and configurable rubric-based lead scoring.
- Developed a **lead-specific AI sales copilot** with RAG context and tool calling for Q&A and personalized email/SMS generation; connected voice agents to CRM actions.
- Built **CRM analytics pipelines and dashboards** for rep activity, lead progression, outreach outcomes, conversion funnels, and sales performance.
- Integrated **Stripe payments and subscriptions** for self-service checkout, billing, and plan upgrades.
- Designed, engineered, and documented the **official Alebex Developer API** and developer accounts for external voice-engine integrations, inbound/outbound calling, and real-time tool calling.

TideSpark | *AI Automation & DevOps Engineer* — Vancouver, BC | **Aug 2025 - Feb 2026**

- Designed and optimized **CI/CD pipelines** for automated build, test, and deployment of microservices and ML models.
- Managed **Docker/Kubernetes infrastructure** for scalable AI workloads, improving resource utilization by **~27%**.
- Built **LLM deployment, prompt API, tuning, and RAG pipelines**, improving relevance by **~30%+** and reducing P95 latency from **~1.2s to ~700ms**.
- Implemented **Elastic Observability** across logs, metrics, and traces for production visibility and debugging.

Operto | *Full Stack Developer & IT Intern* — Vancouver, BC | **Jan 2025 - Aug 2025**

- Developed and maintained **70+ Prefect flows in Python**, reducing manual work for Customer Success/Product teams by **~40%**.
- Triaged production issues across **SQL, REST APIs, JavaScript, and AWS (S3, Cognito)** with end-to-end debugging.
- Compacted legacy **Parquet data in S3**, cutting storage overhead by **~50%** and improving Athena query performance by **~80%**.
- Managed IT operations for **80+ employees**, including MDM, hardware inventory, onboarding, and support.

YMCA BC | *Information Systems Coordinator* — New Westminster, BC | **Sep 2022 - Apr 2023**

- Managed infrastructure and technical operations across **15+ YMCA sites** as part of a six-person IT team.
- Managed **3,000+ users and 2,000+ devices** with Azure AD and Microsoft Intune for identity, compliance, and security policies.
- Resolved **~100-150 support tickets/month** across hardware, software, identity, and network issues while maintaining SLA prioritization.
- Automated **Windows Autopilot provisioning with PowerShell**, reducing device setup from **~40-60 minutes to ~10-20 minutes**.

PROJECTS

Context Snap (JavaScript / Chrome API / Firefox API) — **Mar 2026**

- Built and shipped a **cross-browser MV3 extension** restoring tabs, scroll, form state, and text selections in **<500ms**; published on Chrome and Firefox.
- Engineered a **10-fallback selector system** with dynamic ID detection for reliable restoration across React, Vue, and Angular apps.
- Implemented **framework-aware recovery + progressive DOM traversal** with privacy-first local storage (**0 backend**).

Spotify Auto Curator (TypeScript / Next.js / PostgreSQL / Redis) — **Oct-Dec 2025**

- Built a **dual multi-signal scoring engine** using genre, popularity, skips, and listening history, improving playlist relevance by **~30%**.
- Implemented **Redis + Bull workers** with scheduled jobs/retries, processing listening data at **5-minute intervals**.
- Re-architected candidate discovery after Spotify deprecated Recommendations/Audio Features APIs **without downtime** to existing users.
- Containerized **PostgreSQL 15 + Redis 7 with Docker Compose** on a Raspberry Pi; added Zod, Winston, and Helmet.

Chess Puzzle Generator (TypeScript / Next.js / Prisma / Stockfish) — **Oct-Nov 2025**

- Built and shipped a full-stack app that analyzes Chess.com history and **auto-generates personalized tactical puzzles**.
- Engineered a **client-side Stockfish 17 Web Worker pipeline** at depth 12-15 with UCI parsing/FIFO queuing, enabling **$0 inference cost**.
- Designed **3-class tactical detection**, yielding **~3-8 puzzles/game** while filtering **>80%** of low-quality positions.
- Built **ELO calibration from 1000-2500** using solution length, centipawn magnitude, and theme-weighted difficulty.

SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Java, C#, C++, Go, HTML, CSS
AI / Agents: OpenAI API, RAG, tool calling, LangGraph, HuggingFace, TensorFlow, PyTorch, Prefect
Web / APIs: Node.js, Next.js, React, Express, REST APIs, OAuth 2.0, Twilio Voice/SMS, Nylas API, WebRTC/PSTN
Data / Infra: PostgreSQL, Redis, Prisma, Supabase, MongoDB, Firebase, GraphQL, Docker, Kubernetes, AWS, GCP, Terraform, Elastic, CI/CD
Other: Angular, Flask, .NET, FastAPI, Tailwind, Three.js, Playwright, Vercel, Railway, n8n

EDUCATION

Simon Fraser University | Bachelor of Computer Science, AI Concentration — Burnaby, BC | **Aug 2025**
Dean's Honour Roll | GPA: 3.83/4.00